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ADIKAVI MAHARSHI VALMIKI UNIVERSITY RAICHUR

PROJECT REPORT

ON

"AI POWERED SECURITY SCANNING USING X-RAY IMAGES"

Submitted In Partial Fulfillment of The Requirements

For

VI SEMESTER

BACHELOR OF COMPUTER APPLICATIONS

SUBMITTED BY

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CERTIFICATE

This is to certify that the project entitled **"Ai Powered Security Scanning Using X-Ray Images"** has been successfully submitted by **Ashwini** RegNo. **U23IA22S0017.**

A Bonafide student at Government Degree College, Yadgir in partial fulfillment of award of Bachelor of Computer Applications in the year 2024 – 2025 is record of student own work carried out under my/our guidance. It is certified that all corrections/suggestion indicated for internal assessment have been incorporated in the report & one copy of it being deposited in the Government Post-Graduation Library. The project report has been approved as satisfies the academic requirements in respect of project work prescribed for the said bachelor's degree. It is further understood that this certified the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion draw there in but approved the project only for the purpose for which it is submitted.

Examiners:-

Guide
Name & Guide signature

1.

2.

HOD
Department of Computer Applications

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I express my sincere gratitude to my guide**, Shri. Manjunath. V**, **Assistant Professor** of department of computer applications to guide for their guidance, proper advice & encouragement during the course or work.

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I am also thankful to the entire faculty & staff members and my friends from the computer applications department for their direct & indirect help & cooperation.

Date:- **ASHWINI**

Regno:- U23IA22S0017

CANDIDATE DECLARATION

ASHWINI a student of Bachelor's in Computer Applications department bearing Regno **U23IA22S0017** of **Government Degree College, Yadgir** hereby declare that my full own responsibility for the information, results and conclusions provided in this work titled **"AI POWERED SECURITY SCANNING USING X-RAY IMAGES"** submitted to **Adikavi Maharishi Valmiki University, Raichur** for the award of Bachelor's Degree In Computer Applications.

To the best of my knowledge, this project has not been submitted in part or full elsewhere in any other institution/organization for the award of any certificate/diploma/degree. I have completely taken care of acknowledging the contribution of others in academic work. I further declare that in any case the candidate will be slowly responsible for the same.

Date:-

Place:-

Signature of the candidate

Name:- **ASHWINI**

Regno:- **U23IA22S0017**

Abstract

Security screening using X-ray imagery is a critical task in public safety, especially in airports, government facilities, and high-risk zones. This project presents an automated weapon detection system that leverages deep learning to analyze X-ray images and accurately identify concealed threats such as knives, guns, and other hazardous objects. The system integrates a Convolutional Neural Network (CNN)-based object detection framework, such as YOLOv5 or Faster R-CNN, to localize and classify potential weapons within complex baggage imagery.

The model architecture follows a supervised learning approach, trained on a curated dataset containing annotated X-ray images with bounding box labels for various weapon categories. The dataset is preprocessed with techniques like image normalization, data augmentation, and grayscale conversion to enhance model robustness. Transfer learning is employed using pre-trained backbones to accelerate training and improve detection accuracy in low-data scenarios.

To increase detection precision and reduce false positives, the system incorporates non-maximum suppression (NMS), confidence threshold tuning, and real-time image analysis pipelines. The model's performance is quantitatively evaluated using object detection metrics including Precision, Recall, Intersection over Union (IoU), and Mean Average Precision (mAP).

This AI-powered solution enables rapid, accurate analysis of security images, minimizing human error and enabling faster response times. Applications include intelligent baggage scanning in transportation hubs, border security, and smart surveillance systems. The results highlight the effectiveness of deep learning in enhancing security screening processes, demonstrating its potential in real-world deployment for proactive threat detection and public safety reinforcement.

Keywords: - Deep Learning, X-ray Image Analysis, Weapon Detection, Convolutional Neural Networks (CNN), YOLOv5, Faster R-CNN, Object Detection, Security Screening, Public Safety, Image Preprocessing, Transfer Learning, Mean Average Precision (mAP), Artificial Intelligence, Threat Identification, Computer Vision.

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